

LSTM vs Transformer

Stock Price Prediction — Fair Comparison Report

Fresh inference run: 2026-04-28 11:18

Symbols evaluated: 17 / 17 • Test split: last 15% (chronological)

Avg MAPE → LSTM: 8.82% | Transformer: 8.78%

Avg R² → LSTM: -2.417 | Transformer: -2.147

Avg DirAcc → LSTM: 49.2% | Transformer: 49.4%

Architecture — LSTM:

3 × LSTM layers (128 → 64 → 32 units) + BatchNorm + Dropout + Dense head

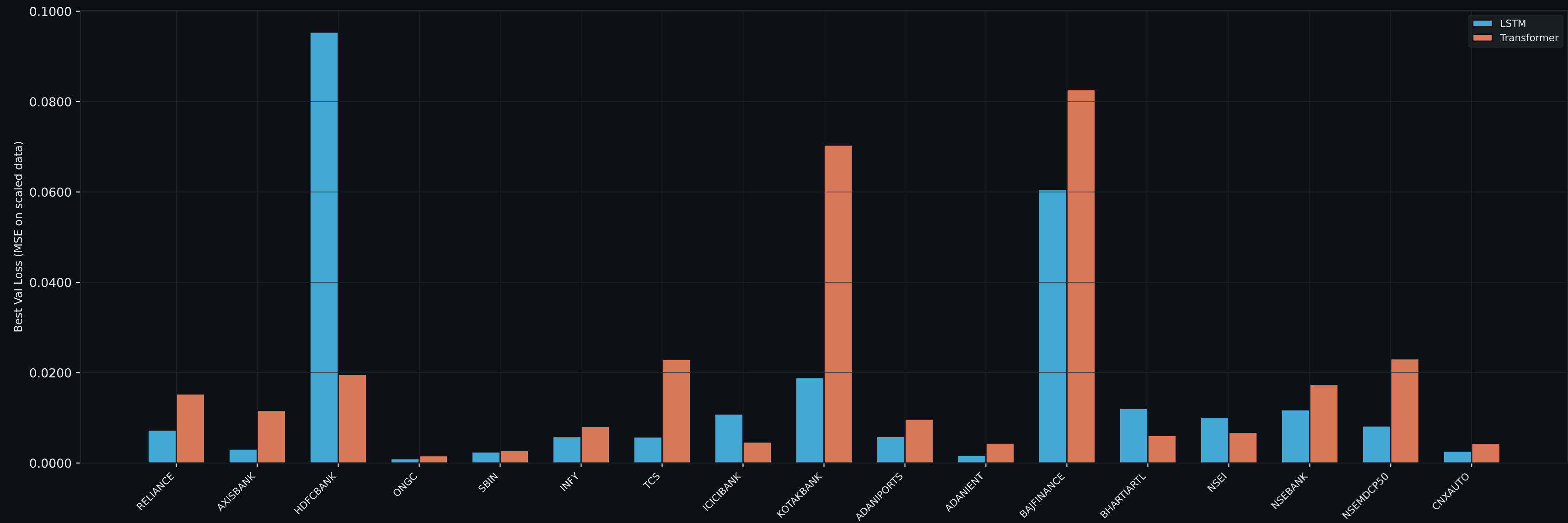
Architecture — Transformer:

3 × Encoder blocks (Multi-Head Attention 8h×64 + GELU FFN 256) + Positional Embedding

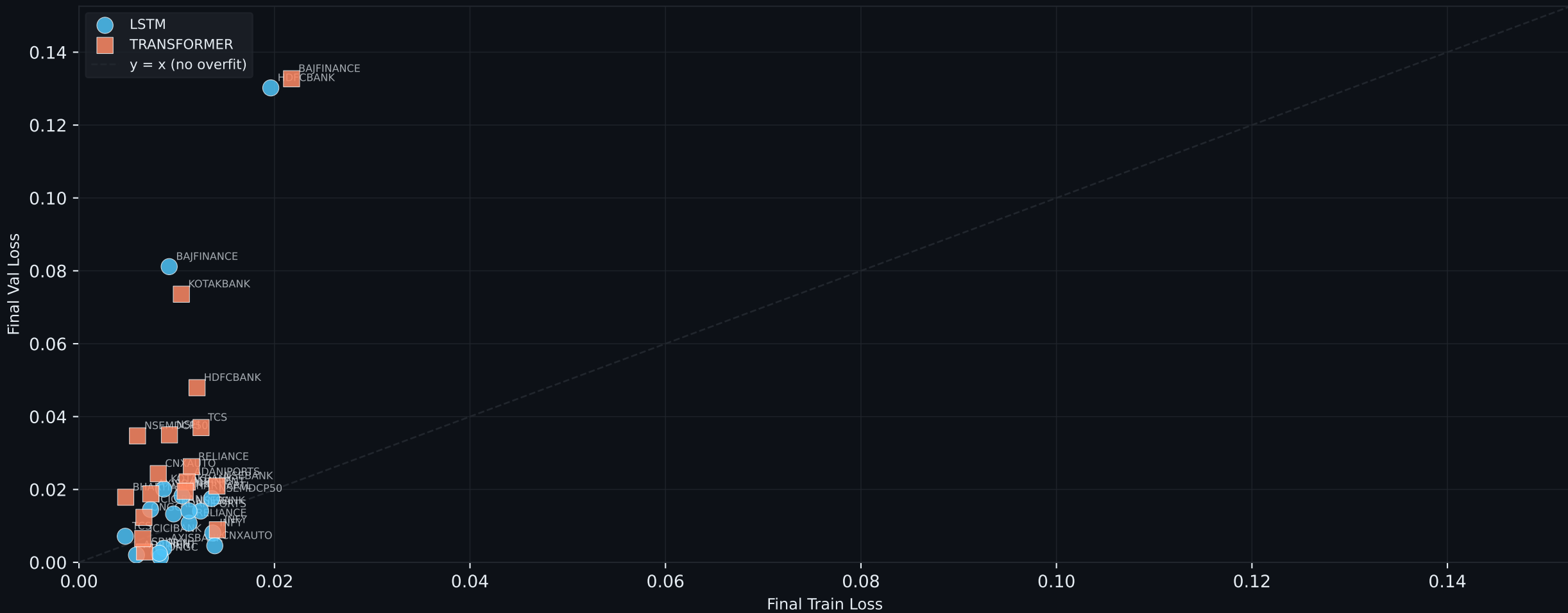
+ GlobalAveragePooling + MLP head (128 → 64 → 1)

Train/Val/Test: 70%/15%/15% chronological • Seq len: 120 • Seed: 42 • Period: 5 years

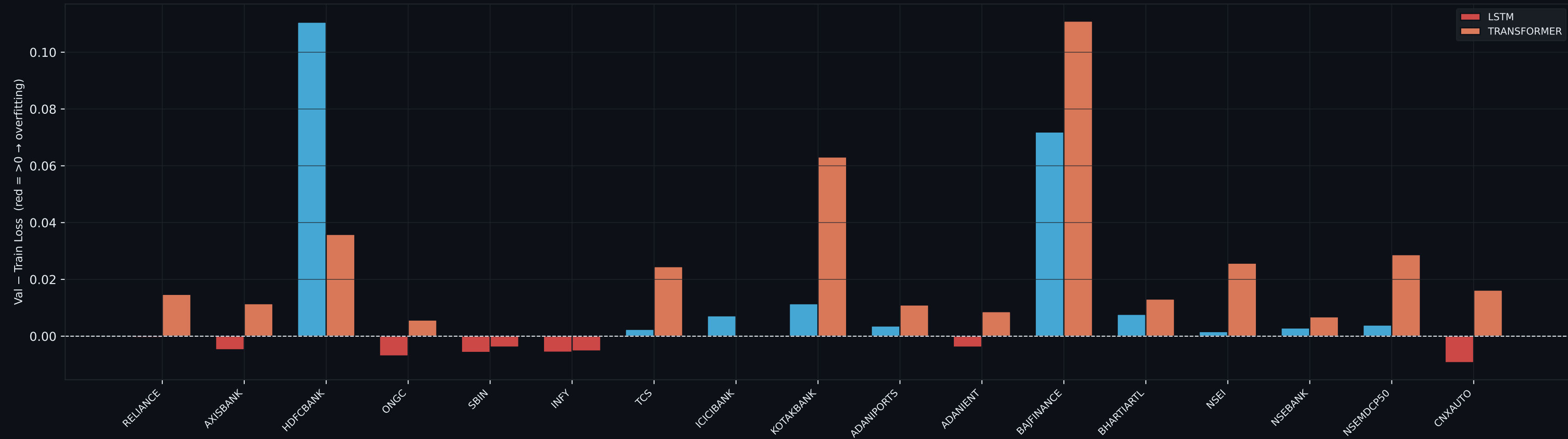
Training Summary — Best Validation Loss per Symbol



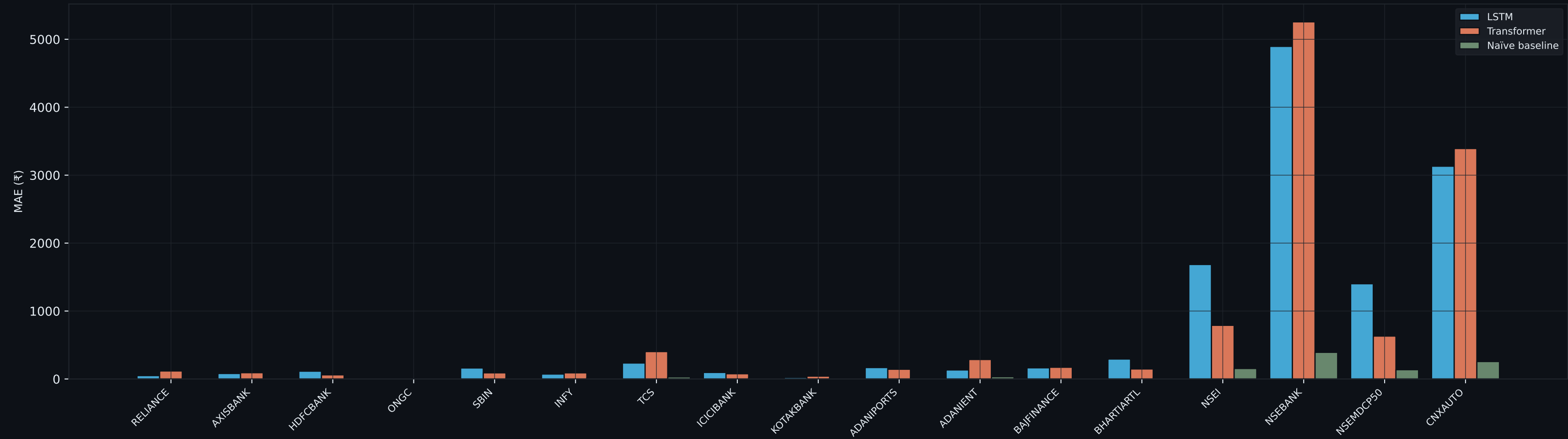
Overfitting Check — Final Train Loss vs Final Val Loss



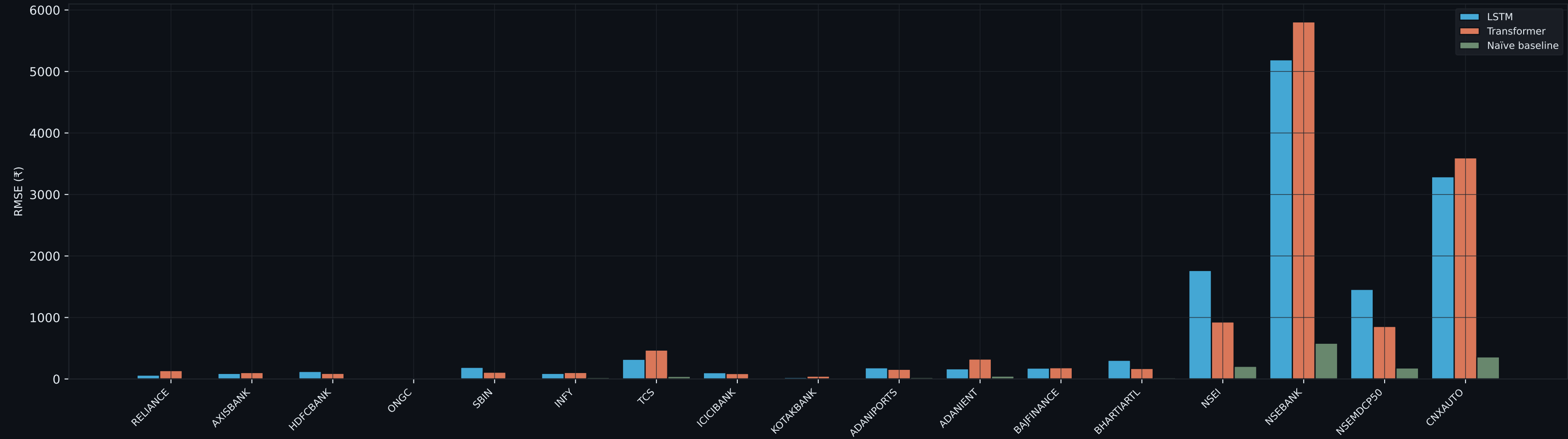
Train-Val Loss Gap (Val – Train) — Overfitting Indicator



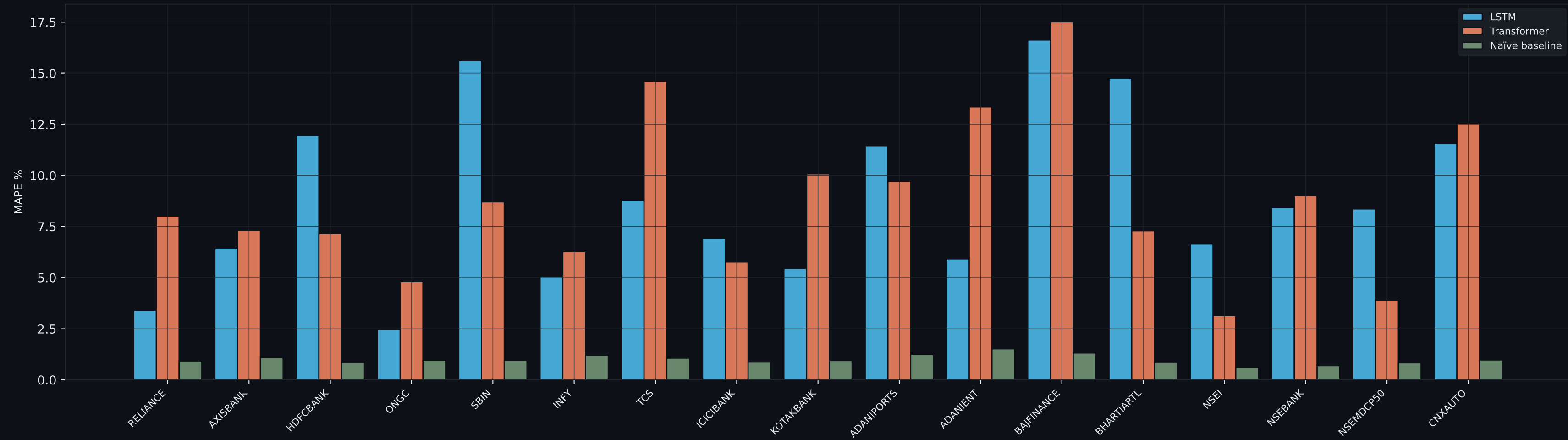
Model Comparison — MAE (₹)



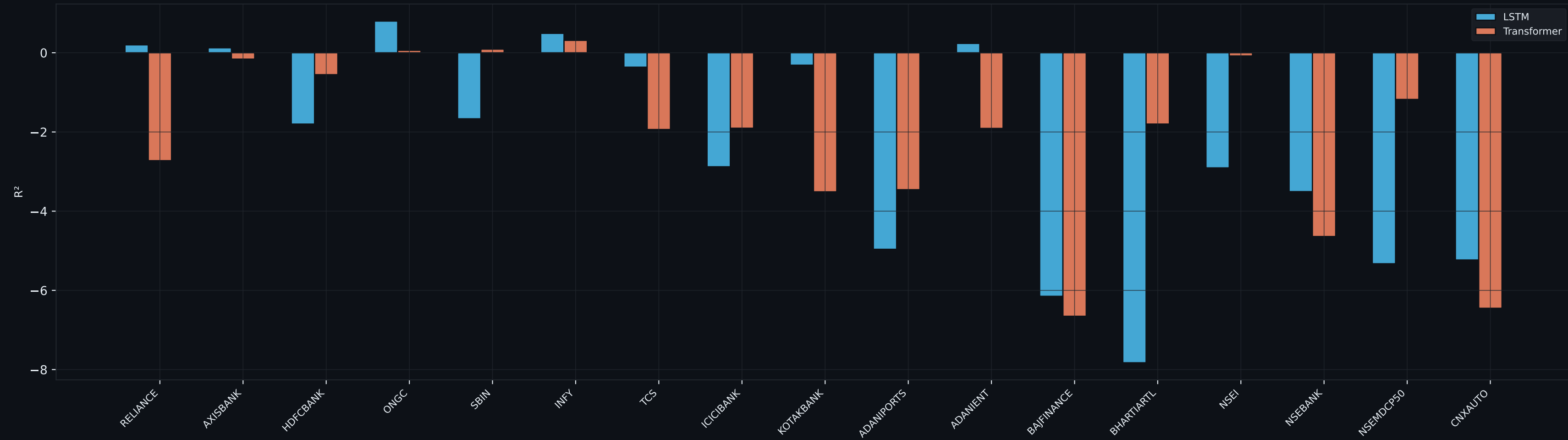
Model Comparison — RMSE (₹)



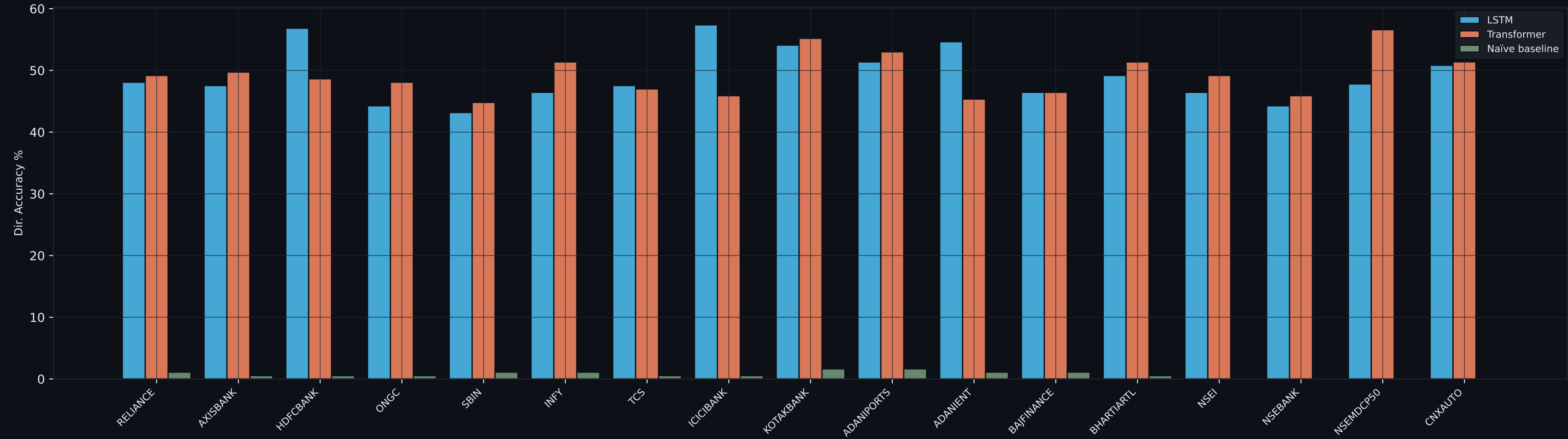
Model Comparison — MAPE %



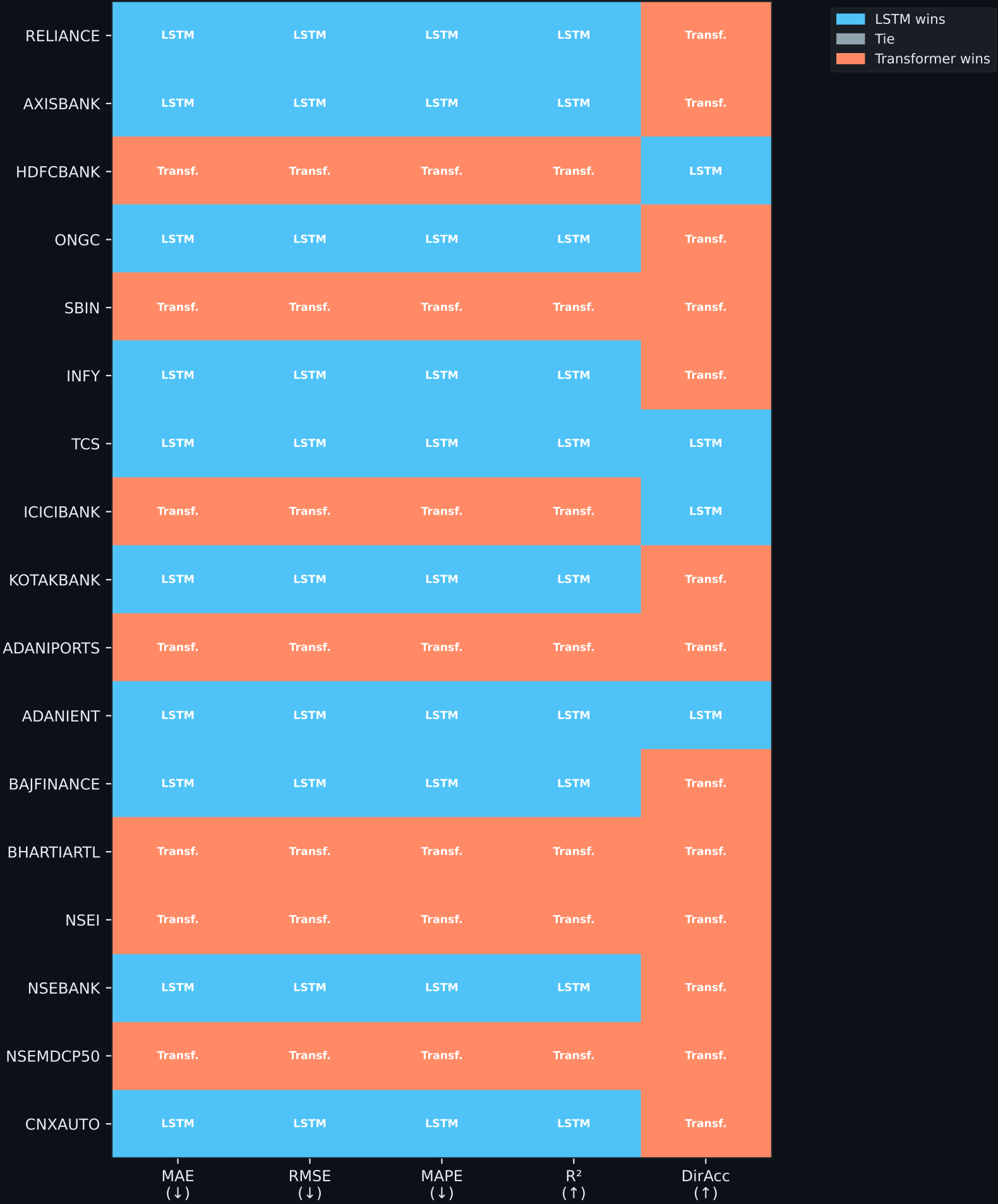
Model Comparison — R²



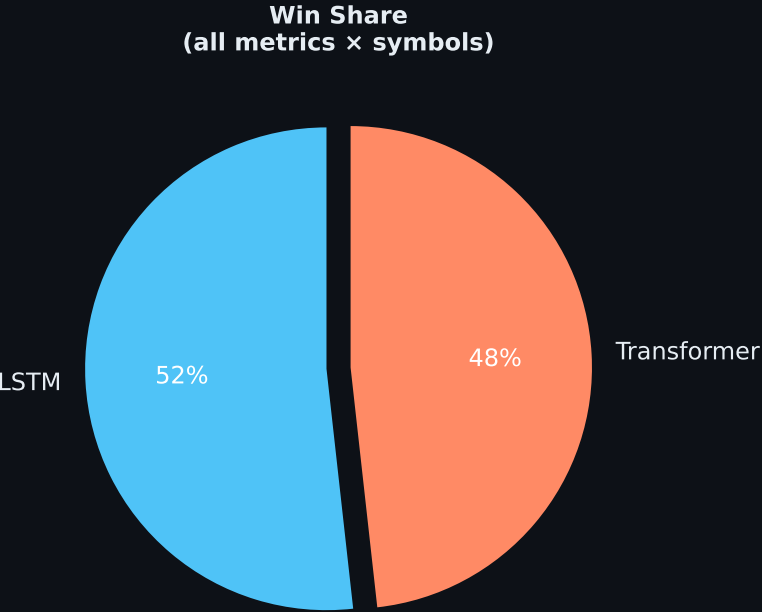
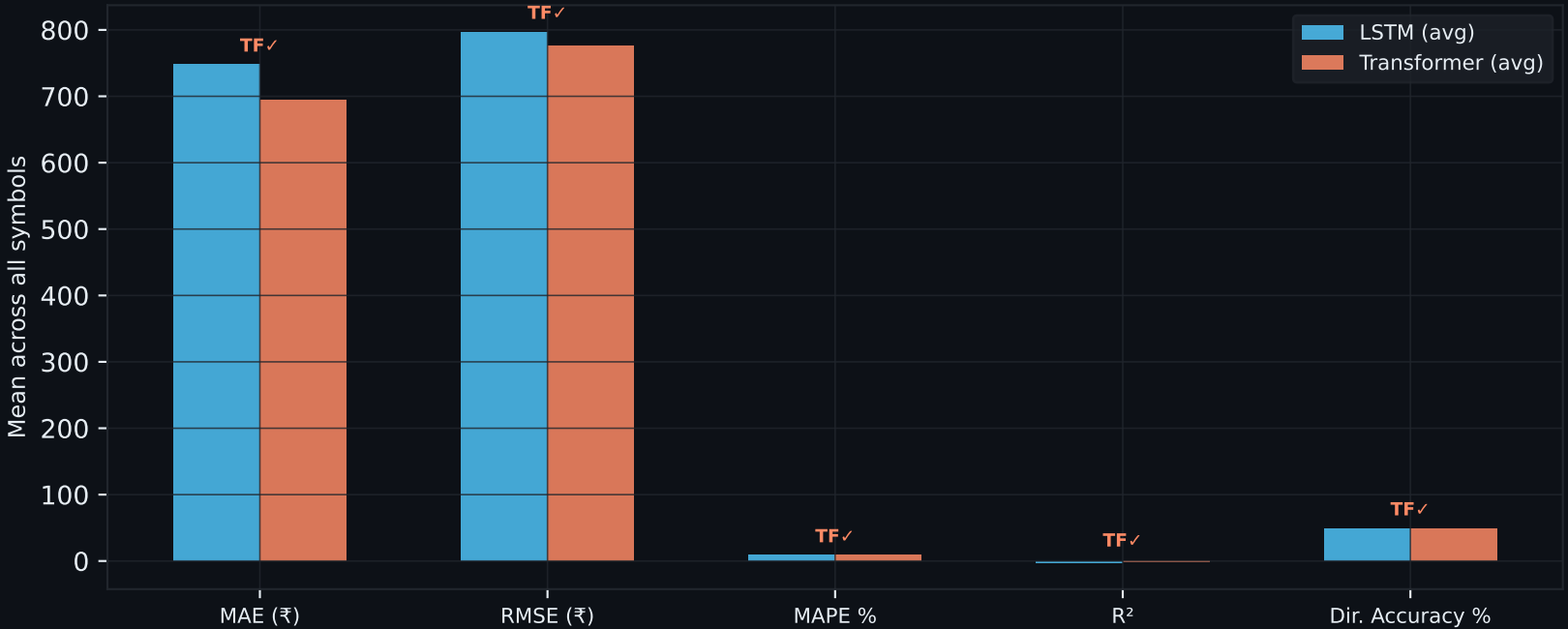
Model Comparison — Dir. Accuracy %



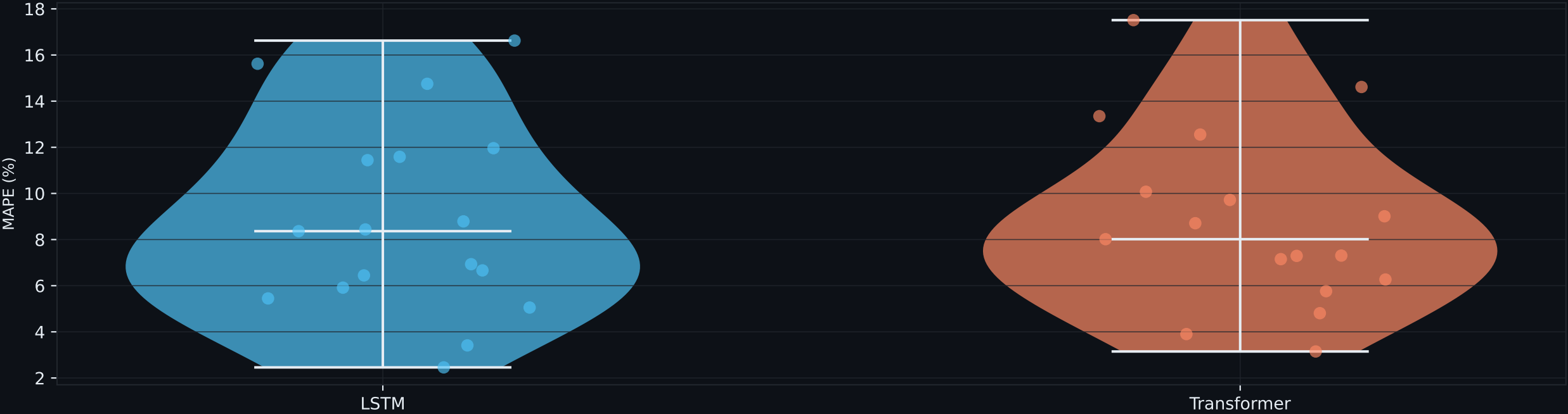
Win Heatmap — LSTM vs Transformer



Aggregate Performance Summary

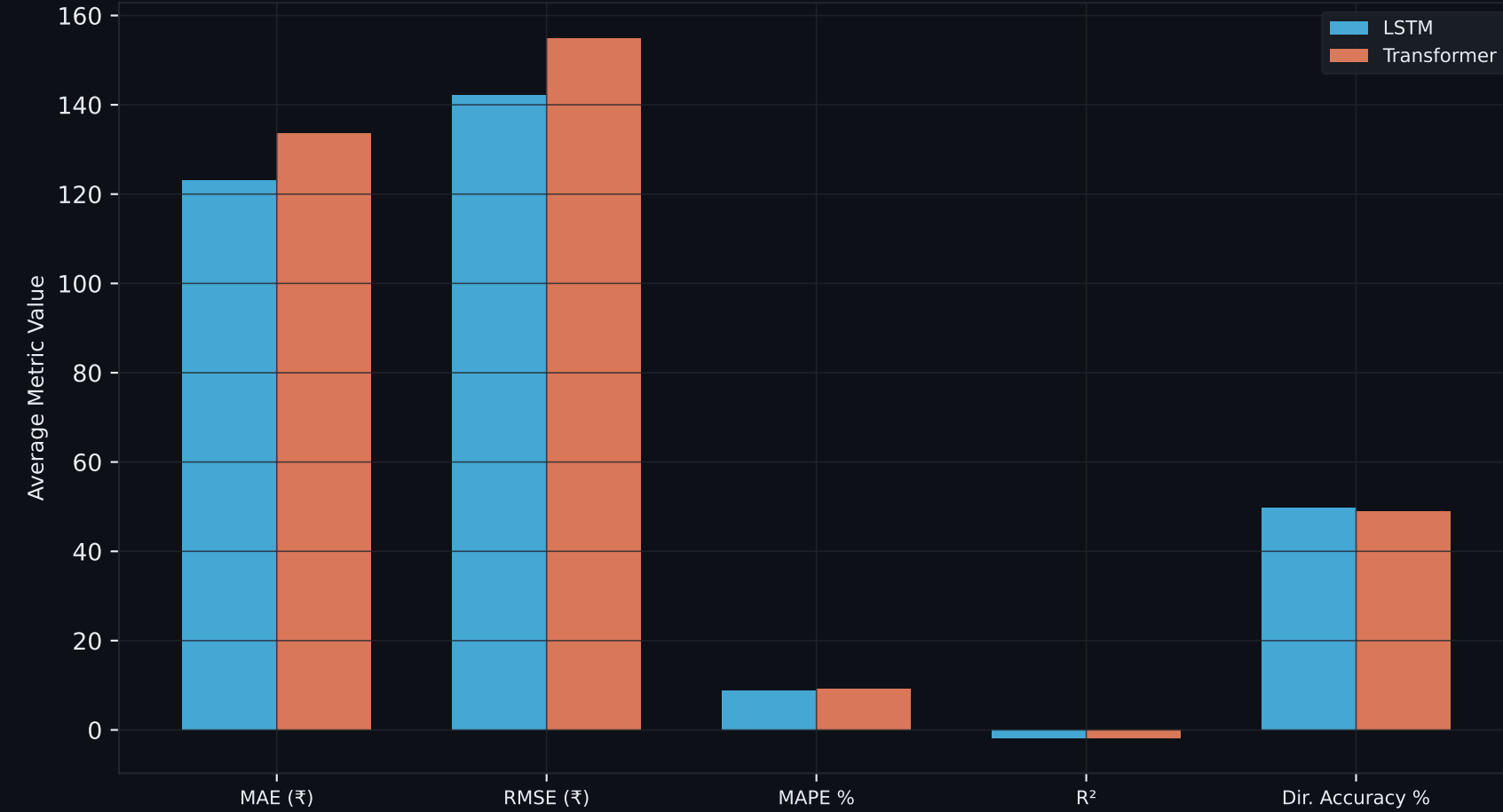


MAPE % Distribution (Test Split)

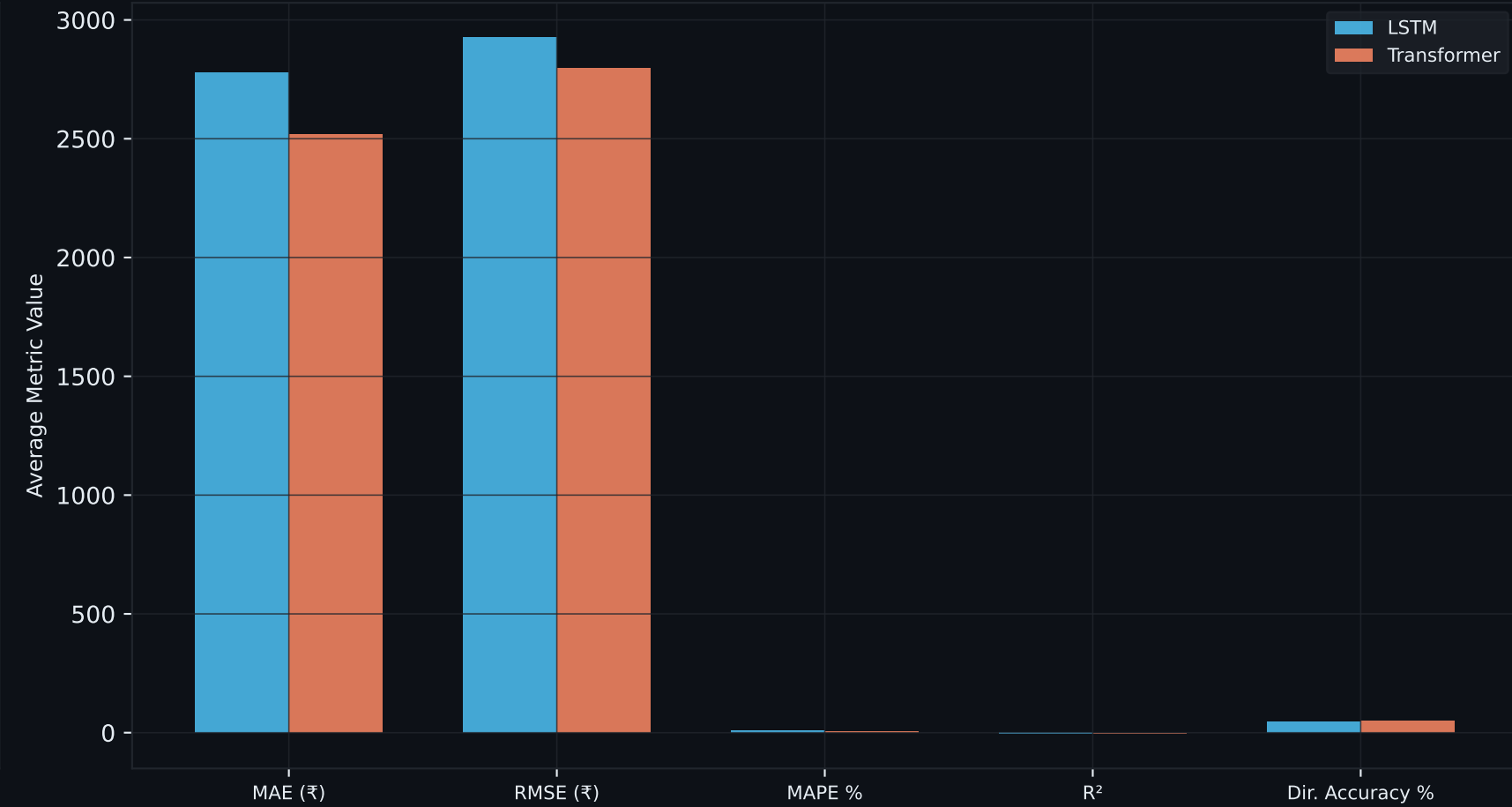


Average Metrics — Stocks vs Indices

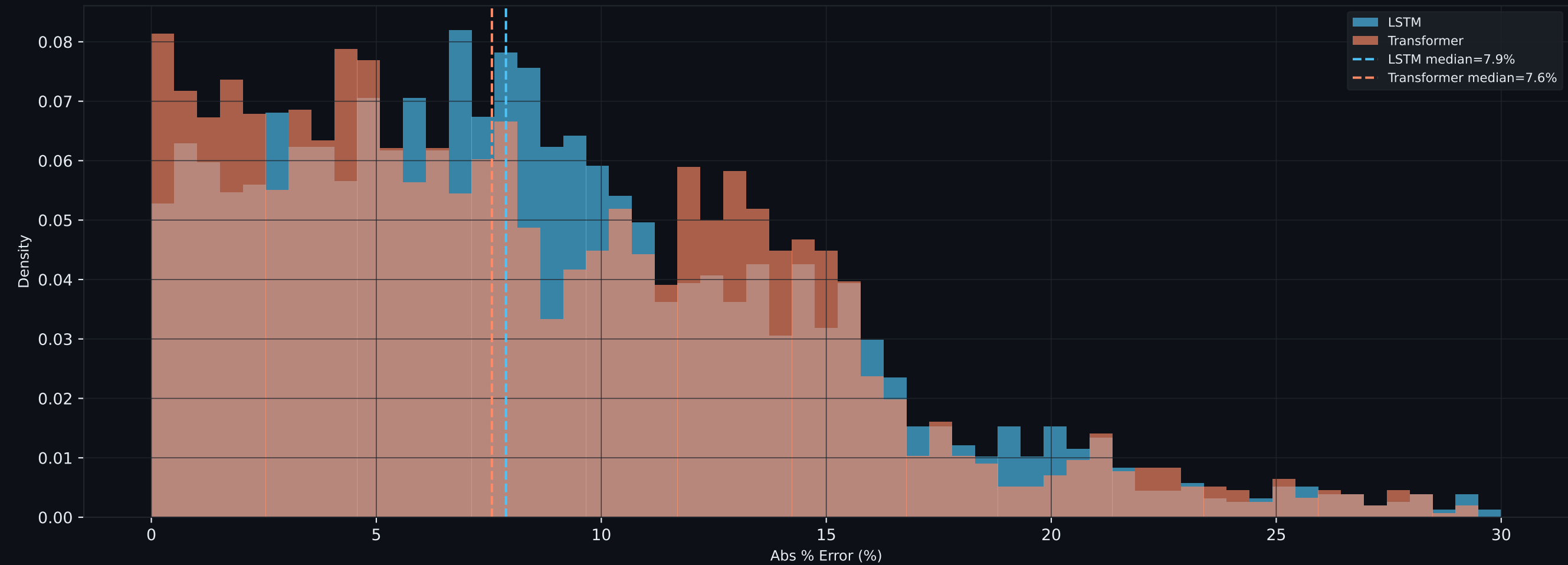
Stocks



Indices



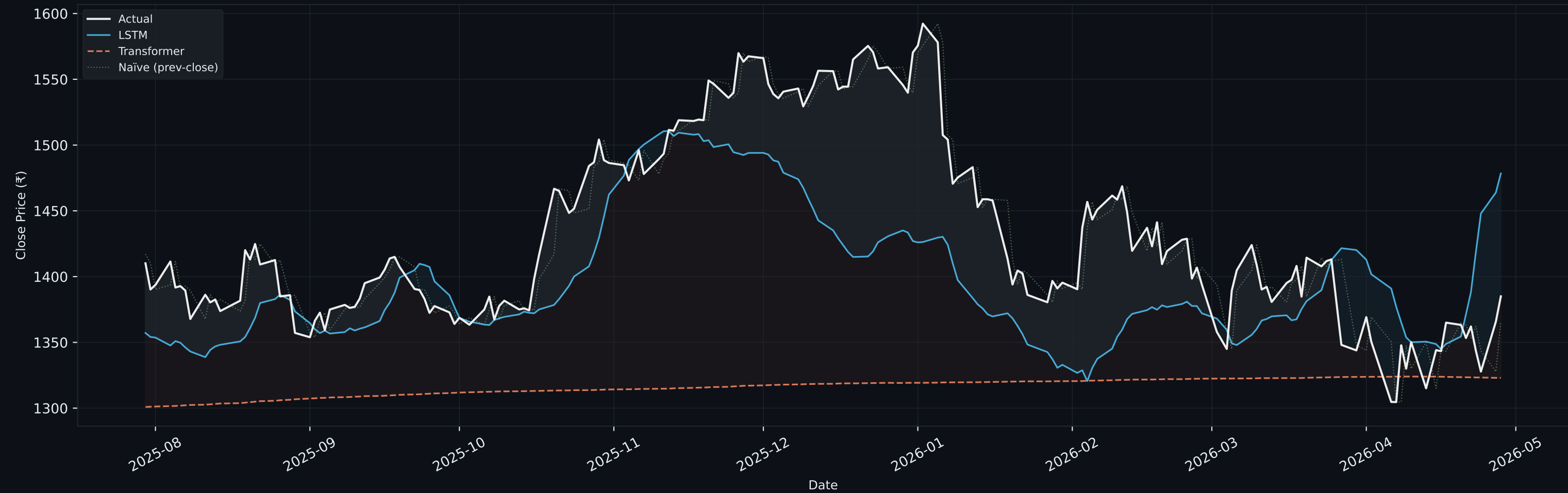
Absolute % Error Distribution Across All Symbols (Test Split)



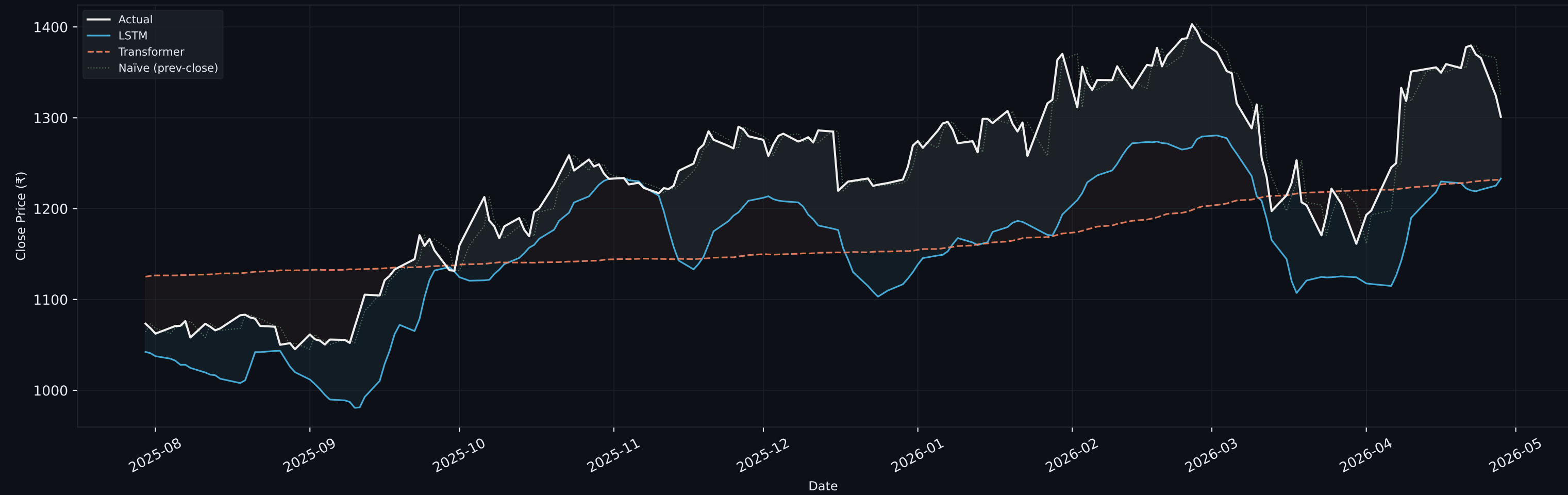
Full Metrics Table — Fresh Inference (Test Split)

Symbol	MAE LSTM	MAE Transf.	RMSE LSTM	RMSE Transf.	MAPE LSTM	MAPE Transf.	R² LSTM	R² Transf.	DirAcc LSTM	DirAcc Transf.	Naive MAE
RELIANCE	49.9	118.0	63.5	137.1	3.41%	8.02%	0.200	-2.725	48.1%	49.2%	13.2
AXISBANK	81.0	92.4	91.3	105.2	6.45%	7.31%	0.125	-0.160	47.5%	49.7%	13.5
HDFCBANK	115.0	60.7	123.6	92.0	11.96%	7.15%	-1.801	-0.553	56.8%	48.6%	7.7
ONGC	6.3	12.8	8.0	17.2	2.46%	4.81%	0.800	0.063	44.3%	48.1%	2.5
SBIN	161.9	90.4	190.2	111.0	15.62%	8.71%	-1.667	0.092	43.2%	44.8%	9.7
INFY	72.5	90.5	91.7	106.4	5.05%	6.27%	0.490	0.314	46.4%	51.4%	17.5
TCS	234.9	403.6	320.8	470.9	8.79%	14.61%	-0.364	-1.938	47.5%	47.0%	30.8
ICICIBANK	96.2	77.3	103.3	89.4	6.93%	5.76%	-2.879	-1.905	57.4%	45.9%	11.8
KOTAKBANK	22.8	42.2	25.4	47.1	5.45%	10.07%	-0.314	-3.512	54.1%	55.2%	3.8
ADANIPORTS	168.3	143.4	183.1	158.3	11.44%	9.72%	-4.963	-3.458	51.4%	53.0%	17.9
ADANIENT	132.9	286.9	166.5	325.0	5.92%	13.35%	0.236	-1.910	54.6%	45.4%	33.6
BAJFINANCE	164.3	172.6	177.8	183.9	16.62%	17.51%	-6.150	-6.654	46.4%	46.4%	12.5
BHARTIARTL	293.7	147.4	304.6	171.6	14.75%	7.29%	-7.828	-1.800	49.2%	51.4%	16.9
NSEI	1685.2	789.8	1765.0	928.1	6.66%	3.15%	-2.907	-0.080	46.4%	49.2%	154.4
NSEBANK	4895.8	5256.7	5190.9	5807.1	8.44%	9.01%	-3.506	-4.640	44.3%	45.9%	392.2
NSEMDCP50	1401.9	631.6	1457.7	855.4	8.36%	3.90%	-5.328	-1.179	47.8%	56.6%	136.7
CNXAUTO	3132.1	3392.4	3289.1	3596.3	11.59%	12.55%	-5.233	-6.452	50.8%	51.4%	256.8

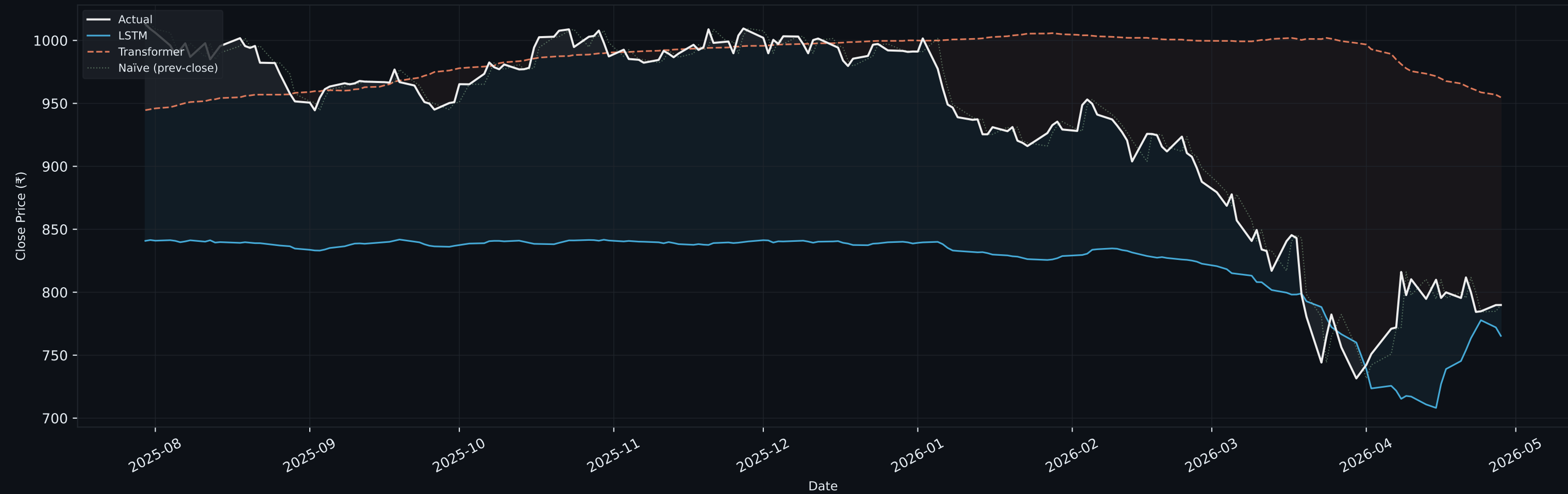
Test-Split Predictions — RELIANCE.NS



Test-Split Predictions — AXISBANK.NS



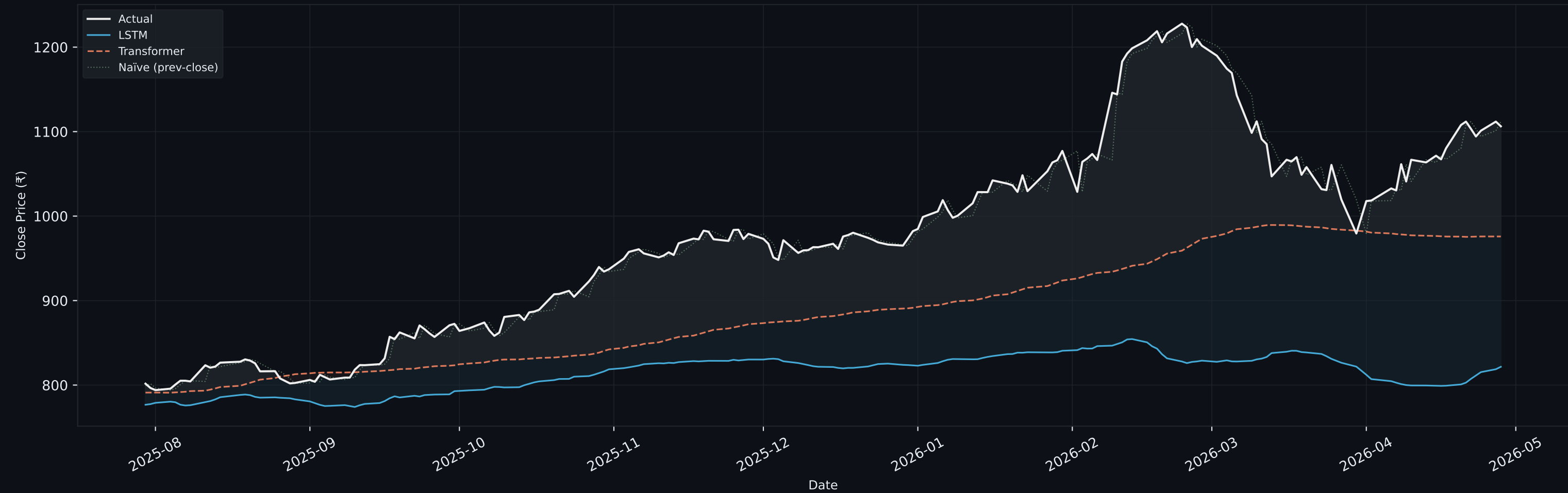
Test-Split Predictions — HDFCBANK.NS



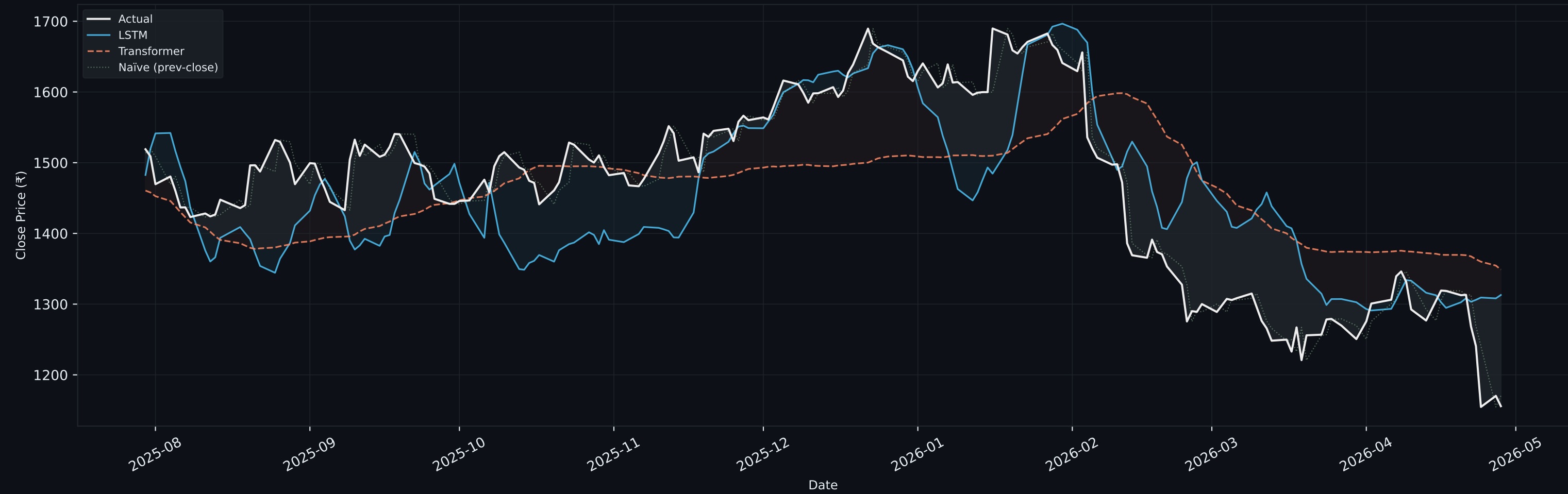
Test-Split Predictions — ONGC.NS



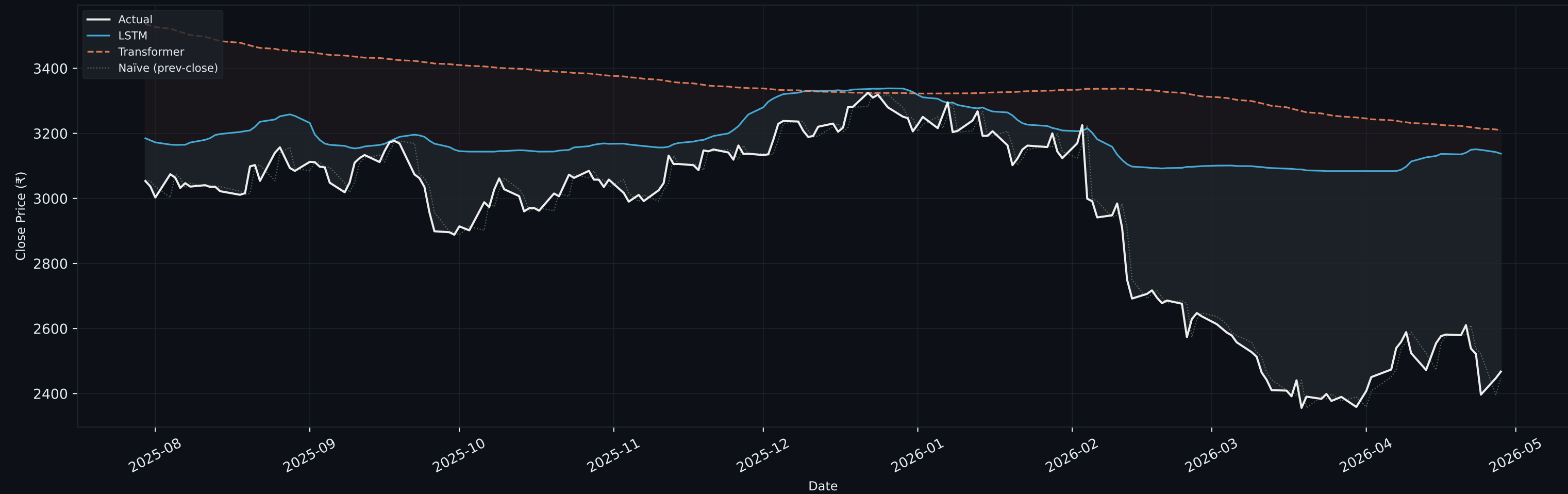
Test-Split Predictions — SBIN.NS



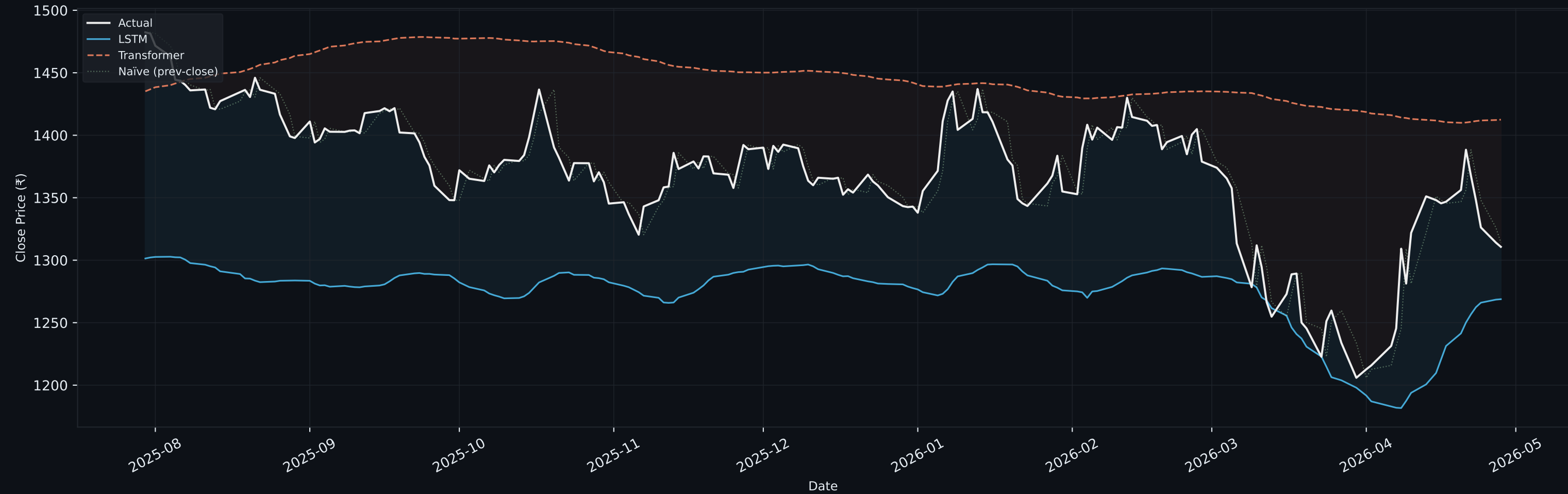
Test-Split Predictions — INFY.NS



Test-Split Predictions — TCS.NS



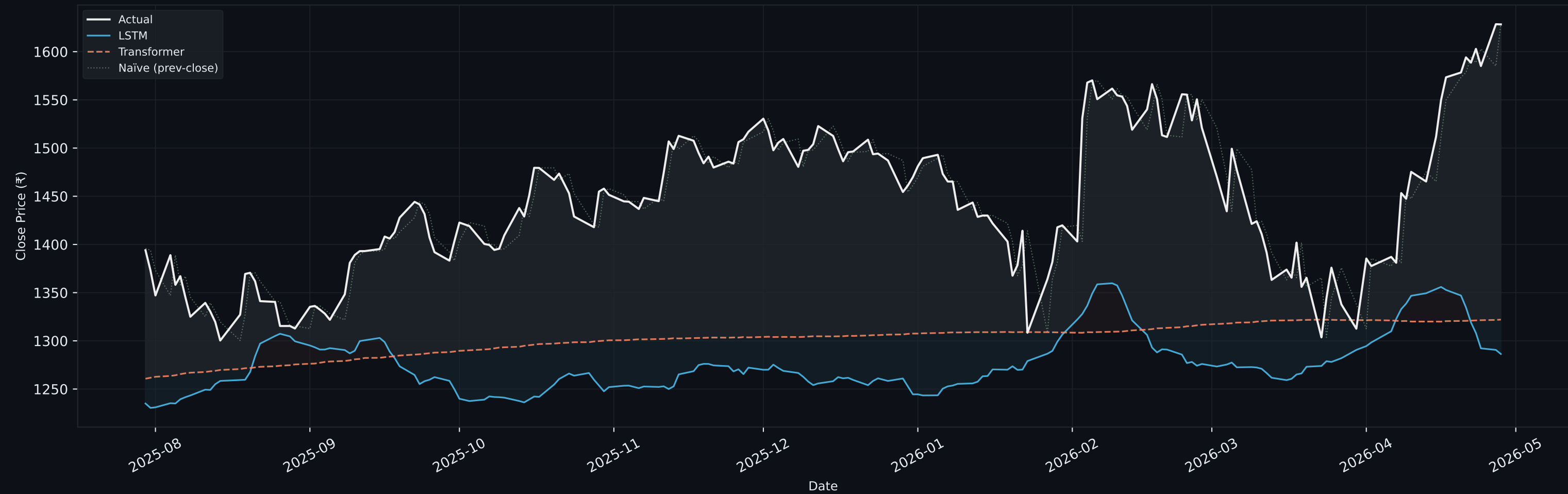
Test-Split Predictions — ICICIBANK.NS



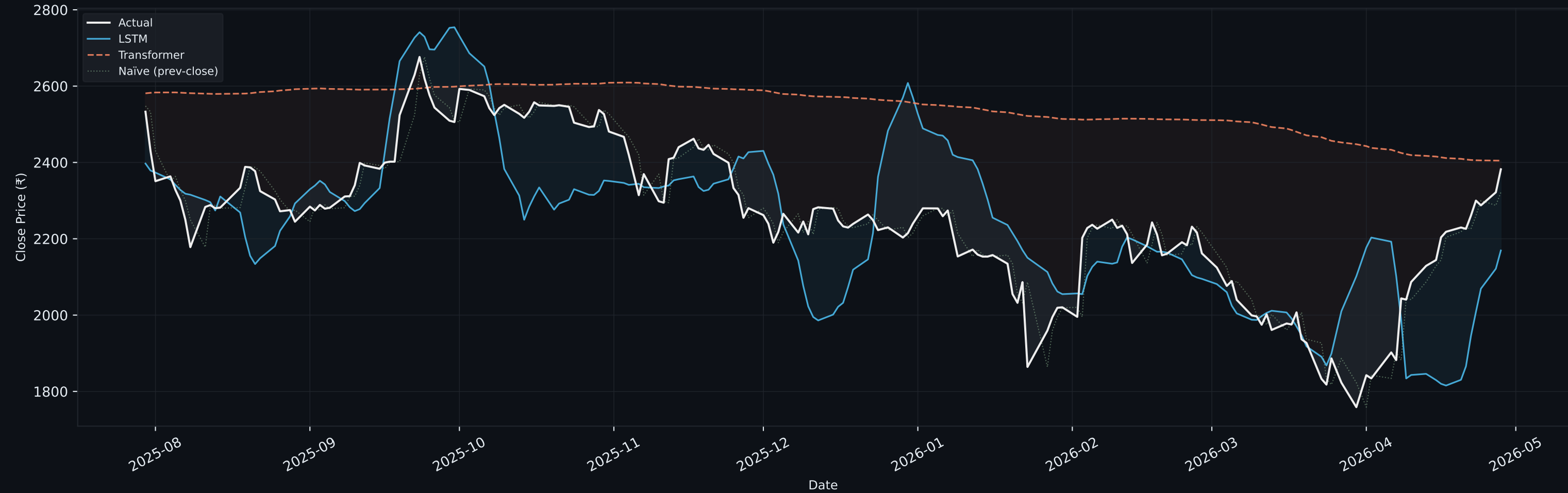
Test-Split Predictions — KOTAKBANK.NS



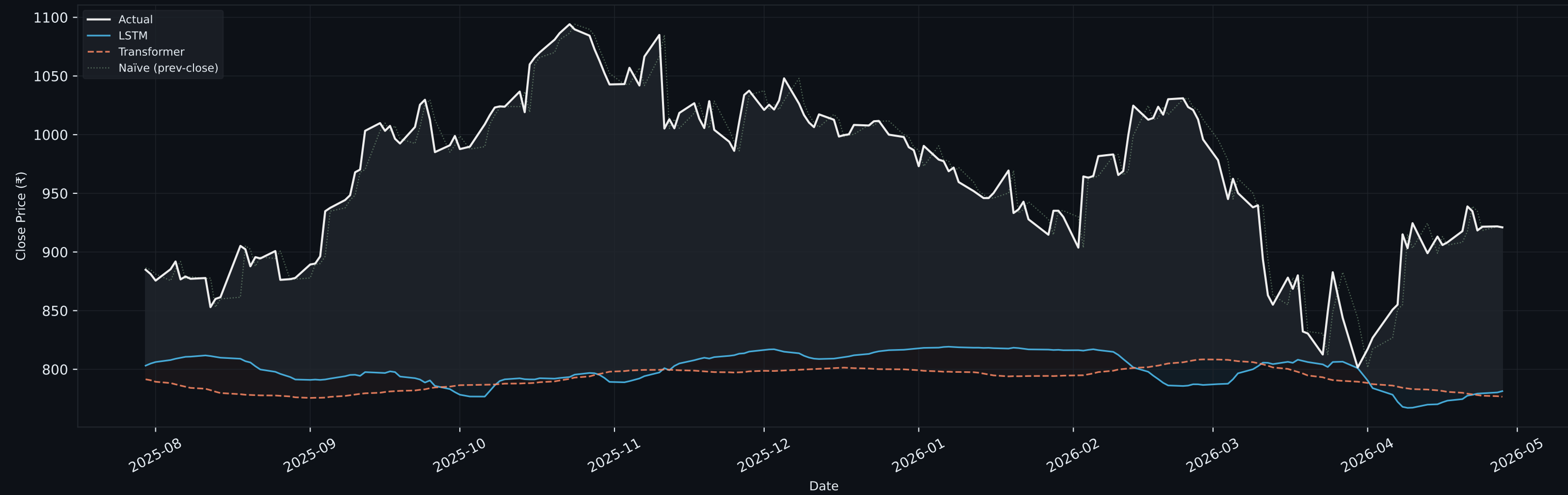
Test-Split Predictions — ADANI PORTS.NS



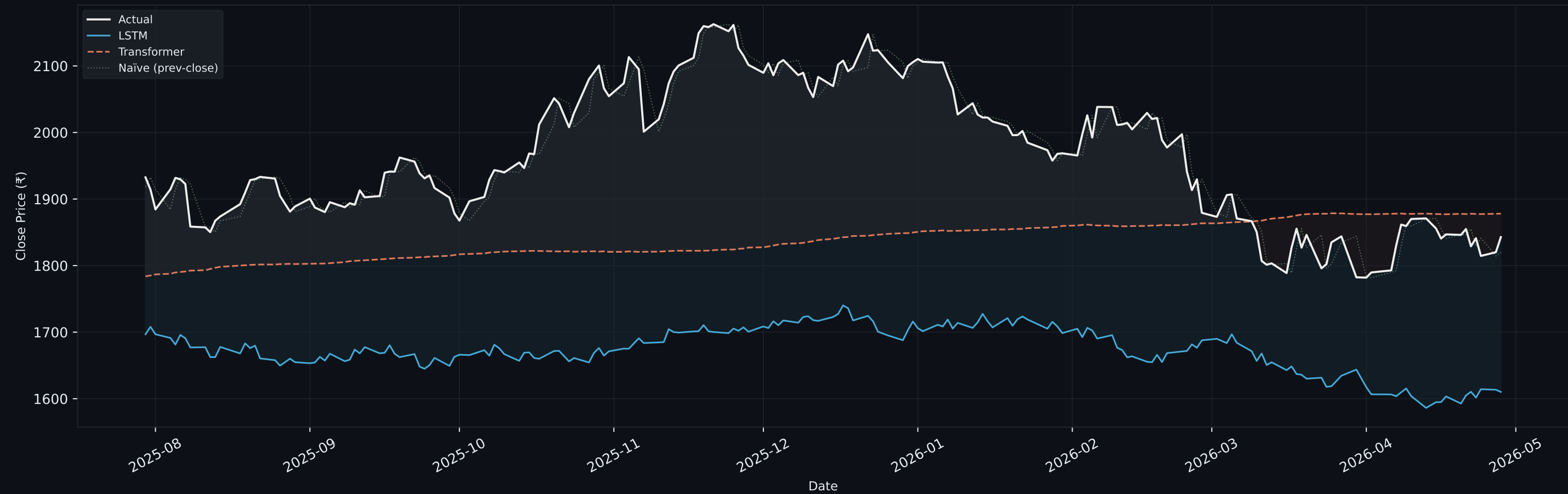
Test-Split Predictions — ADANIENT.NS



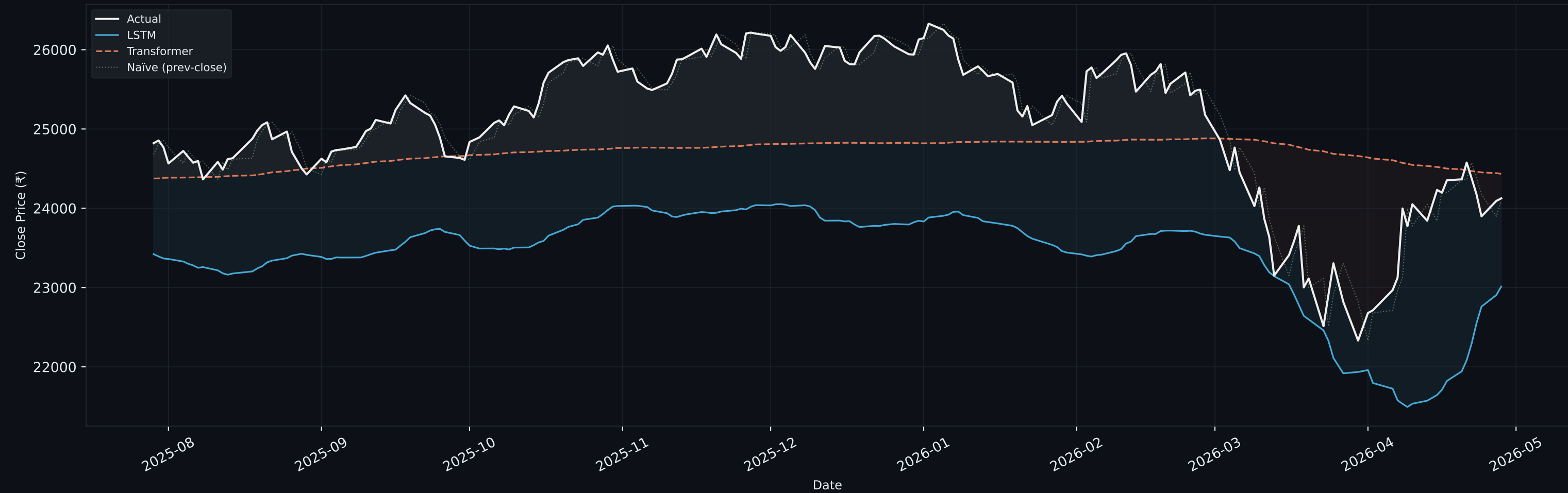
Test-Split Predictions — BAJFINANCE.NS



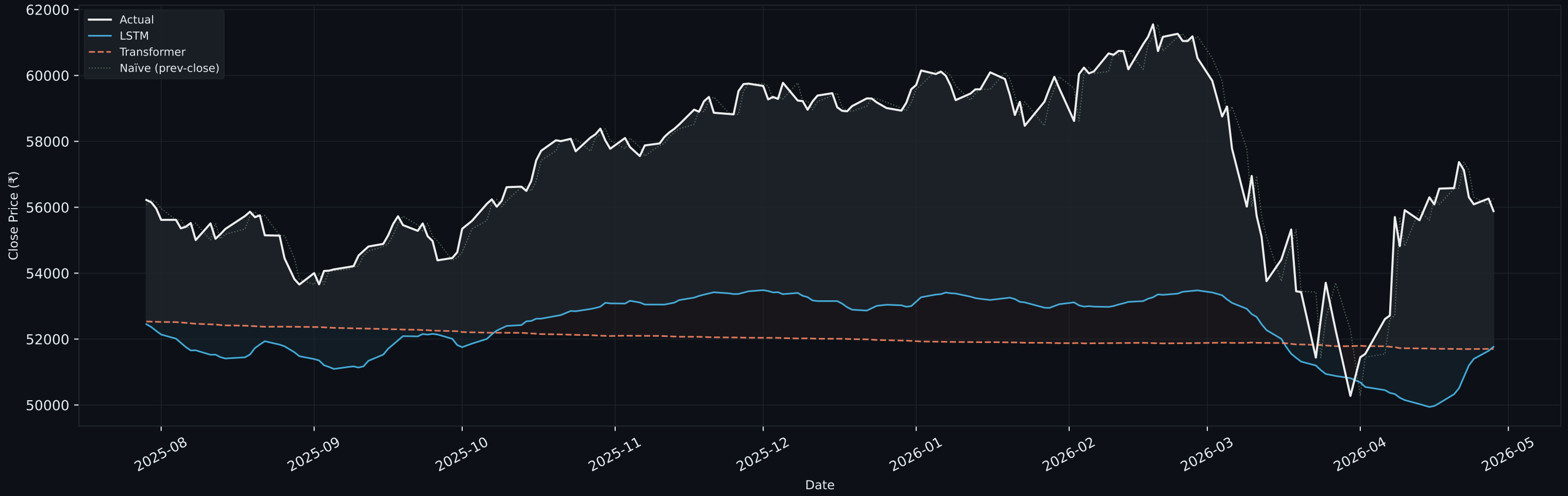
Test-Split Predictions — BHARTIARTL.NS



Test-Split Predictions — $\hat{NSEI}$



Test-Split Predictions — ^NSEBANK



Test-Split Predictions — $\hat{NSEMDCP50}$



Test-Split Predictions — ^CNXAUTO

